

The original dataset for the semester project is the US border crossing dataset that purely focuses on border-crossing stats such as port information and crossing measurement. In order to excavate extra info, two datasets are joined: the GDP stats of each state, and the average temperature of each state, to explore the relation between border-crossing behavior and economical/thermal state.

	Port Name	State	Port Code	Border	Date	Measure	Value	Location	location_x	location_y	year	month	agg_month	avg_temperature
0	Callexico East	California	2507	US-Mexico Border	2019-03-01	Trucks	34447	POINT (-115.48433000000001 32.67524)	-115.48433	32.67524	2019	3	279	48.8 3087
1	Van Buren	Maine	108	US-Canada Border	2019-03-01	Rail Containers Full	428	POINT (-67.94271 47.16207)	-67.94271	47.16207	2019	3	279	24.6 66
2	Otay Mesa	California	2506	US-Mexico Border	2019-03-01	Trucks	81217	POINT (-117.05333 32.57333)	-117.05333	32.57333	2019	3	279	48.8 3087
3	Nogales	Arizona	2604	US-Mexico Border	2019-03-01	Trains	62	POINT (-110.93361 31.340279999999996)	-110.93361	31.34028	2019	3	279	51.7 359
4	Trout River	New York	715	US-Canada Border	2019-03-01	Personal Vehicle Passengers	16377	POINT (-73.44253 44.990010000000005)	-73.44253	44.99001	2019	3	279	29.5 1707

Original dataset

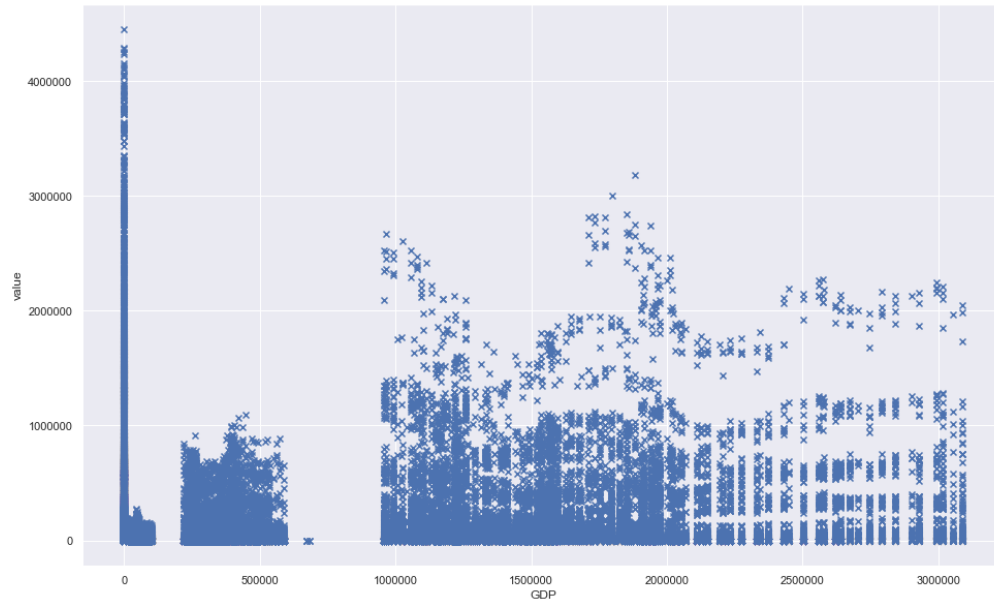
state	2005:Q1	2005:Q2	2005:Q3	2005:Q4	2006:Q1	2006:Q2	2006:Q3	2006:Q4	2007:Q1	2007:Q2	2007:Q3	2007:Q4	2008:Q1	2008:Q2	2008:Q3
Alaska	37377.7	38626	40251.7	42843.8	42633.9	44349.8	45038.2	45719.4	46441.9	48737.4	49602.3	51108.7	52412.2	55933.6	57385
Arizona	219858.1	225360.6	232317	235388.9	242446.7	245035.7	250166.7	255273.5	257952	261341.6	265246	264748.5	262590.9	261815.6	25992
California	1711573.3	1734383.7	1771679.5	1797828.4	1852200	1859177.9	1883269.3	1904301.9	1916188.9	1940770.5	1966137.7	2000327	1971753.2	2012750.2	20198
Idaho	46775.3	47564.9	48675.8	48835.6	50297.2	51074.4	50734.7	51798.8	52797.7	53365.4	54058.5	55186.7	54958.8	55999.5	56171.1
Maine	45234.7	45527.7	45969.6	46767.3	47242.6	47902.8	48108.7	48430.3	48385.9	49154.2	49475.5	49578.8	49706.9	49975.6	50279
Michigan	392832.7	394652.3	393789.4	395641.6	392521.3	395820.3	396592.3	396870.2	404282.4	405322.2	406305.7	399032.8	390804.1	388781.9	38405
Minnesota	238803.4	240523.2	246503.5	245620.7	247639.3	251120.5	249202.5	250977.5	253486.8	258973.1	260097.9	262751	263548.6	264061.2	26861
Montana	29759.4	29951.1	30700.8	31707.5	31827.7	32528.3	33401.6	33975.6	35190.4	36003.3	36274.5	36828.1	36833.9	37196.1	37212
New Mexico	72148.1	73152	73994.8	75441.7	76771.8	77172.1	77569.5	78191.1	78518.9	80243.2	81583.5	81945.6	81702.1	84274.2	84911

GDP by state dataset

Date	Value	Anomaly
199601	24.0	1.9
199602	27.0	0.2
199603	34.8	2.0
199604	41.4	0.3
199605	46.5	-3.0
199606	58.3	1.5
199607	67.0	1.7
199608	64.5	0.8
199609	52.5	-2.4
199610	43.4	-0.8
199611	33.4	1.9
199612	25.3	1.1

Average temperature by state dataset

Both these two datasets are on the grain of each state. After joining these two datasets, the visualization below shows the relation between border crossing values with state GDP.



In general, border crossing values seem to distribute randomly on GDP quantity, with some outliers on low GDP states.

Then look at temperature factor. The below scatter chart plots the border crossing values going with state average temperature. It becomes quite obvious that there are more border crossing records when temperature rises, and the temperature window of [45, 85] locates a large proportion of border crossing activities.

